

CYBERSECURITY CASE STUDY

Centralized Monitoring and SIEM Implementation

Wazuh • Sysmon • File Integrity Monitoring • VirusTotal

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SOC Operations | Detection Engineering | Endpoint Security

Executive summary

This case study demonstrates deployment of a centralized security-monitoring pipeline using Wazuh on Ubuntu and a Windows 10 endpoint. The implementation combined endpoint telemetry, Sysmon process and network events, file-integrity monitoring, and VirusTotal reputation enrichment to support alert triage and threat hunting.

Outcome

A functioning Wazuh 4.12 environment collected Windows telemetry, surfaced Sysmon Event IDs 1 and 3, detected file lifecycle changes, and enriched a controlled EICAR test-file event with multi-engine reputation data.

Role	Individual portfolio case study derived from a collaborative lab environment.
Environment	Ubuntu Wazuh server; Windows 10 monitored endpoint.
Core tools	Wazuh 4.12, Sysmon, Windows Event Log, FIM, VirusTotal.
Security focus	Centralized logging, endpoint visibility, integrity monitoring, enrichment and investigation.
Safety boundary	All activity occurred in an authorized isolated lab using the non-malicious EICAR test file.

Architecture and data flow

The Windows agent forwarded operating-system and Sysmon telemetry to the Wazuh manager. The manager decoded and correlated events, indexed them for investigation, and displayed alerts through the dashboard. FIM generated create/modify/delete events; hash-based VirusTotal enrichment added reputation context for supported file events.

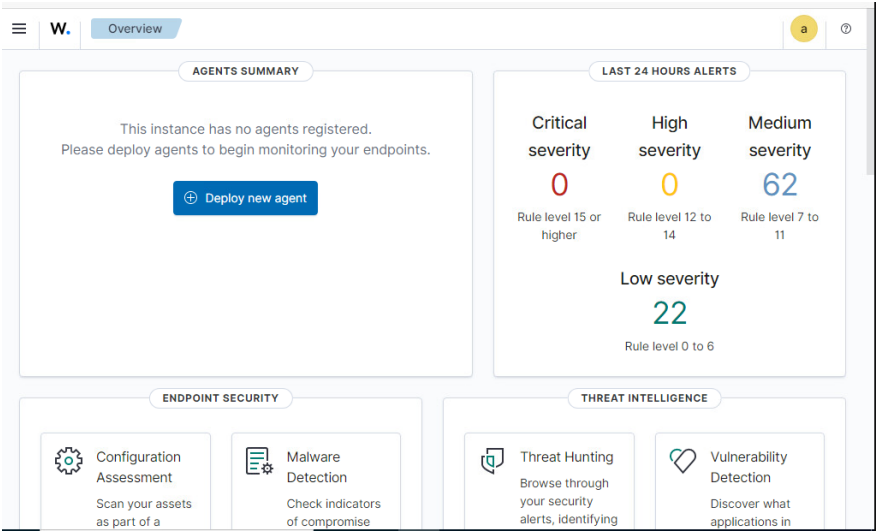


Figure 1. Wazuh overview after server deployment and endpoint enrollment.

Implementation

1. Platform deployment

- Installed and validated the Wazuh manager, indexer and dashboard on Ubuntu.
- Enrolled a Windows 10 agent and confirmed active connectivity from the central console.
- Recorded component versions and service status to establish a reproducible baseline.

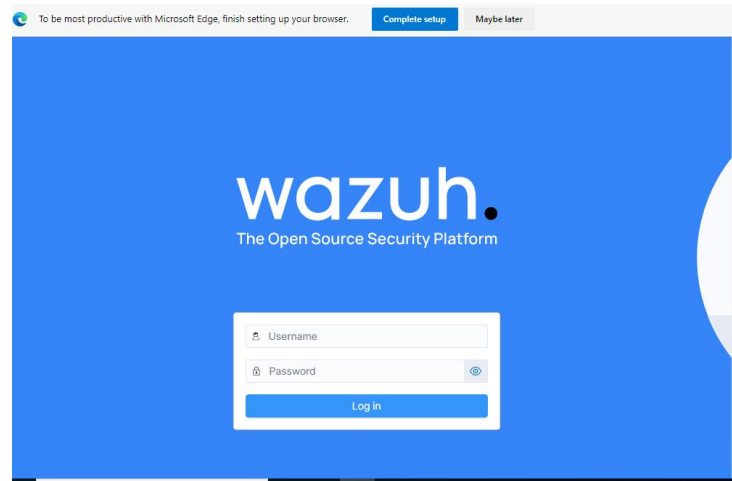


Figure 2. Wazuh dashboard authentication interface confirming web-console availability.

2. Endpoint telemetry with Sysmon

- Installed Sysmon on Windows 10 with a structured community configuration.
- Added the Microsoft-Windows-Sysmon/Operational channel to the Wazuh agent configuration.
- Validated process-creation and network-connection visibility through Sysmon Event IDs 1 and 3.

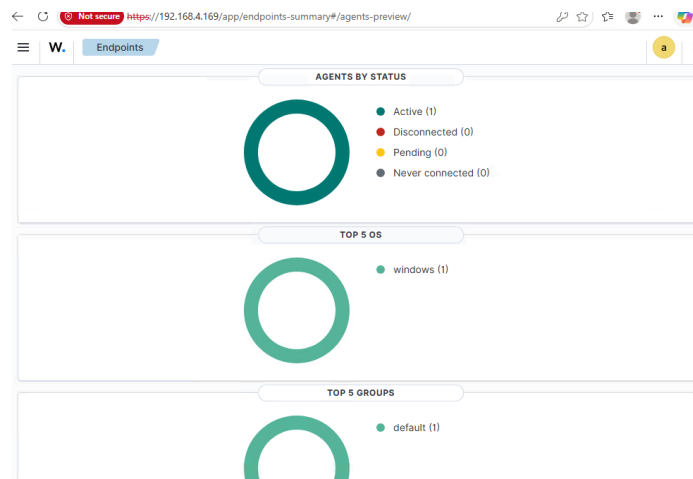


Figure 3. Endpoint and event visibility in the Wazuh dashboard.

Detection validation

File integrity monitoring

A monitored Windows directory was used to create, modify and delete a controlled text file. Wazuh generated integrity events for each lifecycle action, showing how unexpected changes to sensitive paths can be investigated.

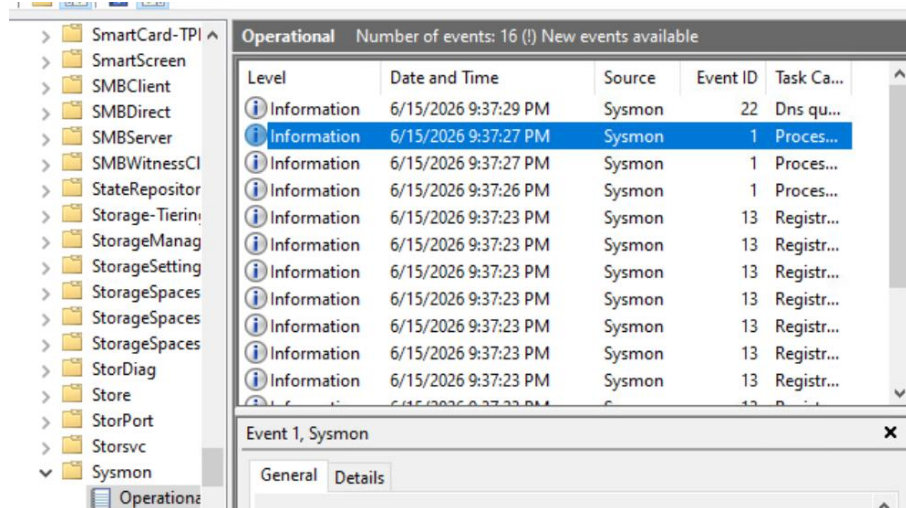


Figure 4. Windows event evidence supporting endpoint-monitoring validation.

Threat-intelligence enrichment

A standard EICAR test file was used to validate enrichment without deploying real malware. The observed hash reputation returned broad antivirus detection, demonstrating how enrichment can help an analyst distinguish a suspicious file from an ordinary integrity change.

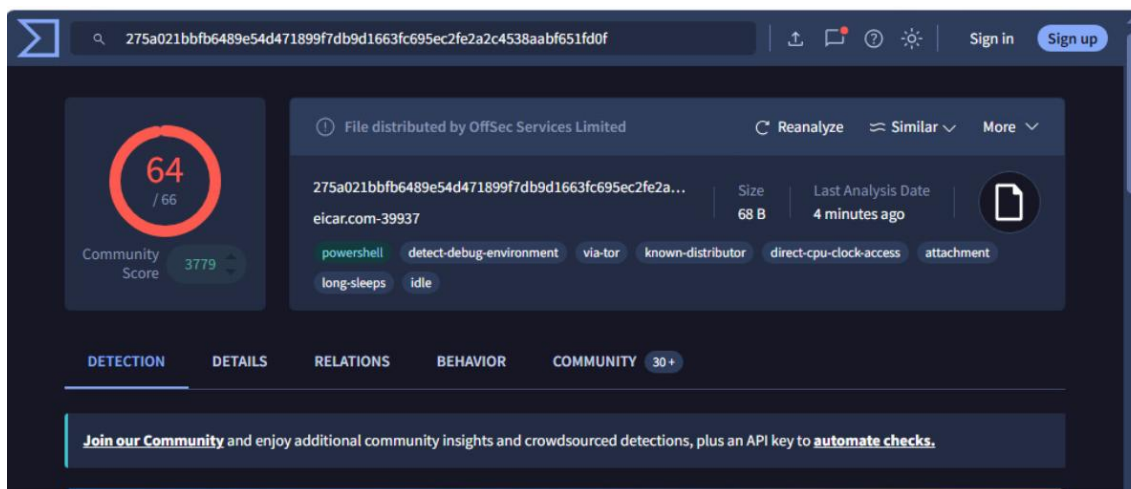


Figure 5. Controlled EICAR reputation result used to validate the VirusTotal workflow.

Security analysis

Analyst value

- Sysmon adds behavioral context beyond standard operating-system logs, including parent process, command line and destination details.
- FIM provides early evidence of unauthorized or unexpected file changes.
- VirusTotal adds external reputation context, reducing the time required for first-pass triage.
- Centralized correlation enables analysts to pivot across host, process, network and file evidence.

Limitations and production improvements

- Reputation is supporting evidence, not a verdict; false positives and unknown hashes still require analyst validation.
- API keys must be protected in secrets storage and never exposed in screenshots or repositories.
- Production deployments should add TLS hardening, least-privilege access, retention policies, time synchronization and tested backup/restore procedures.
- Detection quality should be measured with coverage mapping, tuning, false-positive review and repeatable validation tests.

Skills demonstrated

SIEM engineering	Server deployment, agent onboarding, service validation and centralized event search.
Detection engineering	Telemetry-source selection, controlled event generation and evidence validation.
Endpoint security	Sysmon process/network visibility and Windows event-channel collection.
Threat intelligence	Hash reputation enrichment and analyst interpretation.
SOC analysis	Correlation of process, network, file and reputation evidence.

Conclusion

The project established an end-to-end monitoring workflow from endpoint activity to analyst-visible evidence. The strongest result was not simply generating alerts, but showing how multiple telemetry sources can be correlated to support faster, better-grounded investigation.

Selected references

Wazuh Documentation — <https://documentation.wazuh.com>
Microsoft Sysmon — <https://learn.microsoft.com/sysinternals/downloads/sysmon>
VirusTotal Documentation — <https://docs.virustotal.com/>
EICAR Anti-Malware Test File — <https://www.eicar.org/download-anti-malware-testfile/>